

Before the  
NETWORKING AND INFORMATION TECHNOLOGY RESEARCH AND DEVELOPMENT  
(NITRD) NATIONAL COORDINATION OFFICE (NCO) and  
NATIONAL SCIENCE FOUNDATION  
On behalf of  
THE OFFICE OF SCIENCE AND TECHNOLOGY POLICY (OSTP)  
Alexandria, VA 22314

Request for Comments on the Development of )  
an Artificial Intelligence (AI) Action Plan )      Docket No. NSF\_FRDOC\_0001-3479

**COMMENTS OF CISCO SYSTEMS, INC.**

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## I. INTRODUCTION

Cisco Systems, Inc. (Cisco) submits these comments in response to the *Request for Comments on Development of an Artificial Intelligence (AI) Action Plan* issued by the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO) and the National Science Foundation on behalf of the Office of Science and Technology Policy (OSTP).<sup>1</sup> Cisco is a global provider of networking, security, observability, and collaboration solutions that power the internet and securely connect people. Cisco connects and protects the AI era. With the trusted infrastructure to power and secure AI, Cisco helps maximize AI's value across the economy. Leveraging its industry-leading expertise, Cisco has developed best practices for designing AI-ready infrastructure<sup>2</sup> and has published extensively on AI principles and policy considerations to support the adoption of trustworthy AI. This includes Cisco's Responsible AI Framework,<sup>3</sup> which operationalizes Cisco's Principles for Responsible AI<sup>4</sup> – a set of six core principles guiding our development and use of AI. Cisco is at the forefront of innovation, providing AI-ready networking and data center technologies, AI-enabled cybersecurity, and security solutions to support the trustworthy development and deployment of AI.

As Cisco observed in its recent AI Briefing: CEO Edition, 97% of CEOs say they are planning an AI integration, but only 1.7% report they are fully ready to do so.<sup>5</sup> Similarly, Cisco's AI Readiness Index, found that only 19% of organizations in the United States consider themselves to be AI-ready.<sup>6</sup> These findings indicate that companies may not be prepared to leverage AI to its full potential. Infrastructure limitations, security concerns, and skills and knowledge gaps are the main barriers that are often cited.<sup>7</sup> While there are undoubtedly challenges to unlocking the benefits of AI across the U.S. economy, there are clear actions which the U.S. Government and private sector can partner on to address these barriers and sustain American leadership in AI. Cisco welcomes the opportunity to assist NITRD and OSTP in identifying actions that strengthen AI innovation, encourage secure deployment of AI, and support U.S. workers.

**A summary of recommendations is provided on the last page of this submission.**

## II. INFRASTRUCTURE NEEDED FOR AI GROWTH

Robust digital, physical, and energy infrastructure is necessary to develop and deploy AI across all sectors of the U.S. economy. Cisco plays a pivotal role in AI infrastructure by providing scalable

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<sup>1</sup> Request for Information on the Development of an Artificial Intelligence (AI) Action Plan, 90 Fed. Reg. 9088, (Feb. 6 2025) <https://public-inspection.federalregister.gov/2025-02305.pdf>

<sup>2</sup> Cisco Systems, Inc., *Cisco Data Center Networking Blueprint for AI/ML Applications* (March 2024), <https://www.cisco.com/c/en/us/td/docs/dcn/whitepapers/cisco-data-center-networking-blueprint-for-ai-ml-applications.html>.

<sup>3</sup> Cisco Systems, Inc., *Cisco Responsible Artificial Intelligence Framework* (2024), [https://www.cisco.com/c/dam/en\\_us/about/doing\\_business/trust-center/docs/cisco-responsible-artificial-intelligence-framework.pdf](https://www.cisco.com/c/dam/en_us/about/doing_business/trust-center/docs/cisco-responsible-artificial-intelligence-framework.pdf).

<sup>4</sup> Cisco Systems, Inc., *Cisco Responsible Artificial Intelligence Principles* (2024), [https://www.cisco.com/c/dam/en\\_us/about/doing\\_business/trust-center/docs/cisco-responsible-artificial-intelligence-principles.pdf](https://www.cisco.com/c/dam/en_us/about/doing_business/trust-center/docs/cisco-responsible-artificial-intelligence-principles.pdf).

<sup>5</sup> Cisco Systems, Inc., *Cisco Study: CEOs Embrace AI, but Knowledge Gaps Threaten Strategic Decisions and Growth* (Feb. 2025), <https://newsroom.cisco.com/c/r/newsroom/en/us/a/y2025/m02/cisco-study-ceos-embrace-ai-but-knowledge-gaps-threaten-strategic-decisions-and-growth.html>.

<sup>6</sup> Cisco Systems, Inc., *AI Readiness Index* (2024), [https://www.cisco.com/c/m/en\\_us/solutions/ai/readiness-index.html](https://www.cisco.com/c/m/en_us/solutions/ai/readiness-index.html).

<sup>7</sup> Cisco Systems, Inc. *supra* note 5.

and secure solutions that support the increasing demands of AI workloads.<sup>8</sup> Cisco's AI-native infrastructure is designed to enhance data center operations, offering integrated systems that simplify AI deployment and management. These solutions are part of Cisco's broader strategy to connect people everywhere to the internet and the digital tools they need to succeed.

*A. Support investments in and buildout of data centers, high-speed connectivity infrastructure, and high-performance networks*

Digitization of the global economy and growing adoption of data-intensive technologies like AI require infrastructure that can securely hold and quickly transfer vast amounts of data. This infrastructure includes data centers, components like routers, switches, processors, storage devices, and servers, as well as systems for power and cooling. U.S. public and private sector organizations need to ensure they are equipped with infrastructure that can keep pace with increasing AI workloads. As OSTP evaluates policy measures to support AI innovation, it is critical to recognize the importance of AI-ready data centers and connectivity infrastructure to enhance the performance of AI applications and enable innovation and scalability.

Data centers – from the hyperscale to the modular level – are critical to support AI innovation and adoption. Cisco is supporting AI data centers through our accelerated compute, network fabric, and hyperscale fabric solutions. The U.S. Government can play a role in data center growth by creating an enabling environment that encourages private capital to flow more easily into data center projects. The U.S. Government can streamline regulatory barriers, use government procurement as a market signal, and educate the public about the essential role data centers play in the U.S. economy.

It will also be critical for the U.S. Government to consider the burgeoning influence of edge AI and the proliferation of IoT devices. Edge AI, by processing data closer to its source, significantly alters traditional AI workload distribution, shifting processing from centralized data centers to distributed edge devices. This shift not only impacts network traffic patterns but also introduces novel cybersecurity challenges.<sup>9</sup> Policymakers should consider how to integrate edge AI into the broader AI infrastructure strategy. Such integration is vital to ensure that the U.S. AI infrastructure remains resilient and adaptable in the face of these emerging technological trends.

Networks must have the right hardware, software, and configuration to facilitate the rapid exchange and processing of data and prevent bottlenecks that would impede AI training efficiency and use of AI applications in real-time. High-throughput and low-latency networks help to avoid packet drops which can disrupt the deployment of time-sensitive AI applications.<sup>10</sup> Overall, AI models need scalable and reliable network connectivity. While centralized networking can support the massive requirements for AI training, a more distributed network transport model will be

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<sup>8</sup> With products like the Cisco UCS X-Series Modular System and Nexus 9000 Series Switches, Cisco ensures high-performance computing and networking capabilities essential for AI applications.

<sup>9</sup> With millions of IoT devices generating and processing data at the edge, the attack surface could expand exponentially, necessitating robust security protocols and real-time threat detection capabilities. Scalable solutions at the network's edge, like Cisco's Secure DDoS Edge Protection on routers, can detect and mitigate attacks in real time, keeping attack traffic off the network. See Cisco Systems, Inc., *Cisco Secure DDoS Edge Protection Technical White Paper* (Jan. 25 2023), <https://www.cisco.com/c/en/us/products/collateral/security/secure-ddos-protection/secure-edge-protection-tech-wp.html>.

<sup>10</sup> Cisco Systems, Inc. *supra* note 2

required for real-time inferencing applications – often occurring closer to the end user and data sources.<sup>11</sup> AI will continue to transform the flow of traffic across networks. Downstream traffic is expected to increase as AI applications often require substantial data inputs from cloud-based services to deliver personalized and real-time responses to users. Upstream traffic is also likely to increase due to AI-driven devices collecting and transmitting data for processing and analysis. Overall increases in downstream and upstream traffic will require greater bandwidth. Cisco has developed our Agile Services Networking solution to enhance the deployment and performance of AI applications by optimizing and modernizing network infrastructures to meet the demand for high-bandwidth and secure connectivity necessary for AI.

Current U.S. connectivity infrastructure may be ill-equipped to handle this rising demand. According to Cisco’s 2024 AI Readiness Index, 46% of U.S. respondents acknowledged their infrastructure has limited or moderate scalability and flexibility to accommodate the increasing demands from AI.<sup>12</sup> As the volume and complexity of AI data flows increase, it will be critical for the U.S. Government to help address challenges related to legacy infrastructure. Outdated infrastructure is more vulnerable to security threats and can be an inhibitor to AI adoption as legacy systems often have performance limitations and lack the capacity to handle the data and the high-speed processing that AI applications require. Public and private sector organizations will need to consider how to identify, prioritize, and replace equipment that has reached its end of life.

Reliable internet access is vital for deploying AI tools and ensuring their accessibility to end-users. High-speed, low-latency broadband is key to realizing the return on investment from AI technologies, as it enables individuals and businesses to leverage AI tools effectively in real-time. However, more than 6 million Americans lack high-speed broadband access,<sup>13</sup> and the challenge is exacerbated in rural areas.<sup>14</sup> Broadband access has been shown to boost the productivity of workers and businesses and to improve market access, strengthening economic competitiveness.<sup>15</sup> Cisco is focused on improving the cost of broadband deployment<sup>16</sup> and supports the government’s efforts to close the digital divide and connect all Americans to vital internet resources.

Additionally, one of the major challenges broadband providers face is the complex and lengthy permitting requirements. These barriers can delay the expansion and enhancement of network infrastructure. By addressing these barriers, the government can accelerate broadband deployment, ensuring that connectivity infrastructure keeps pace with technological advances.

Ensuring that all Americans have access to AI technology will help strengthen and grow the U.S. economy. We encourage OSTP to promote policies that enable investment in data centers,

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<sup>11</sup> Rakesh Chopra, *A Service Provider Architecture for AI-Ready Connectivity*, Cisco Blogs (Feb. 10 2025) <https://blogs.cisco.com/sp/a-service-provider-architecture-for-ai-ready-connectivity>.

<sup>12</sup> Cisco Systems, Inc., *supra* note 6.

<sup>13</sup> Federal Communications, Commission, *FCC National Broadband Map* (Feb. 4 2025), <https://broadbandmap.fcc.gov/area-summary>.

<sup>14</sup> Federal Communications, Commission, FCC Fact Sheet, 2024 Section 706 Report, GN Docket No. 22-270 (Feb. 22 2024), <https://docs.fcc.gov/public/attachments/DOC-400675A1.pdf>.

<sup>15</sup> Bert Kroese, *Can Internet Access Lead to Improved Economic Outcomes?*, World Bank Blogs (April 5, 2022), <https://blogs.worldbank.org/en/digital-development/can-internet-access-lead-improved-economic-outcomes>.

<sup>16</sup> Cisco Routed PON, which simplifies service providers’ end-to-end architecture, helps to lower operating expenses. See Bill Gartner, *Transforming the Economics of Superfast Broadband with Cisco Routed PON*, Cisco Blogs (March 15 2024), <https://blogs.cisco.com/sp/transforming-the-economics-of-superfast-broadband-with-cisco-routed-pon>.

broadband infrastructure, and the development of future-proofed networks. To this end, the U.S. Government should:

- Encourage strategic investments in new data center capacity as well as modernization of legacy data centers;
- Promote best practices for building AI-ready networks;<sup>17</sup>
- Ensure federal broadband investments are implemented efficiently and quickly to ensure all Americans have access to AI tools; and
- Work with permitting agencies to streamline regulatory requirements to expedite the construction of data centers and the deployment of high-speed broadband to all Americans.

*B. Ensure data centers have reliable and affordable energy and energy efficient technology*

Modern data centers for AI applications require reliable energy to power complex processing operations and to provide adequate cooling systems.<sup>18</sup> However, current energy infrastructure, built mainly in the 1960s and 1970s “is struggling to meet our modern electricity needs,” according to the U.S. Department of Energy.<sup>19</sup> Moreover, the electricity grid needs the capacity to accommodate more distributed and variable energy sources and to withstand extreme weather impacts. Aging infrastructure also carries a higher risk of failure that could negatively impact local communities. Given the establishment of data centers as critical infrastructure, ensuring data centers have reliable and affordable energy is challenge government and industry must work together to meet.

However, with this challenge comes an opportunity to modernize the electricity grid and embed energy efficiency within data center operations to ensure the U.S. can meet future power demands. An energy grid equipped with digital solutions (such as Cisco’s Distribution Automation and Advanced Metering Infrastructure technologies) provides smart monitoring and energy management capabilities for intermittent and distributed generation to ensure efficient and reliable power delivery, advanced analytics to manage load shedding and peak shaving, and automated data collection to reduce maintenance costs.<sup>20</sup>

Data centers can also reduce energy costs and maximize resources with energy efficient technology.<sup>21</sup> The Federal Energy Management Program (FEMP), notes that ICT equipment “can account for over half of the entire facility’s energy usage” and therefore recommends use of energy efficient equipment, which can “significantly reduce these loads,” and help boost Power Usage Effectiveness, a key efficiency metric for data center operations.<sup>22</sup> The federal government should promote the use of energy efficient IT equipment in data centers (e.g., routers, servers, and processors) and complementary OT to help reduce energy consumption and lower the costs from

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<sup>17</sup> Cisco Systems, Inc. *supra* note 2.

<sup>18</sup> Electricity and water are critical resources for the power subsystems, as well as uninterruptible power supplies, ventilation and cooling systems, fire suppression, and backup generators in data center facilities.

<sup>19</sup> U.S. Department of Energy Grid Deployment Office, *What Does it Take to Modernize the Electric Grid?* (Oct. 19, 2023), <https://www.energy.gov/gdo/articles/what-does-it-take-modernize-us-electric-grid>.

<sup>20</sup> Cisco Systems Inc., *Digitize the Power Automation Grid Reliably, Safely, and Efficiently* (2023), <https://www.cisco.com/c/dam/en/us/solutions/global-partners/cisco-se-grid-modernization-so-r2.pdf>.

<sup>21</sup> See Cisco Systems Inc., *Comments of Cisco Systems Inc.*, National Telecommunications and Information Administration and U.S. Department of Energy Request for Comments on Bolstering Data Center Growth, Resilience, and Security (Docket No. NTIA-2024-0002) (Nov. 4 2024), <https://www.regulations.gov/comment/NTIA-2024-0002-0027>.

<sup>22</sup> Otto Van Geet & David Sickinger, FEMP, *Best Practices Guide for Energy-Efficient Data Center Design* (2024), <https://www.energy.gov/sites/default/files/2024-07/best-practice-guide-data-center-design.pdf>.

AI workloads. This should be a priority for all new data center capacity but there is also an opportunity to reap the benefits of energy efficient technology by modernizing legacy systems.

U.S. energy infrastructure development is crucial as AI compute demands outpace addition of new generation to the grid. The U.S. Government should adopt a dual approach that accelerates energy acquisition but also enhances the efficiency of technology infrastructures supporting AI. Initiatives that enhance the efficiency of data center hardware and software, optimize algorithms for reduced computational load, and foster research into computing paradigms that minimize energy consumption should be prioritized. Simultaneously, strategic investments in grid modernization and diverse energy sources are crucial to ensure reliable power delivery to support an AI-driven economy. To strengthen the resiliency of U.S. energy grid for AI and to achieve the objectives outlined in President Trump's Executive Order 14154,<sup>23</sup> the U.S. Government should:

- Encourage investment in grid modernization through the use of grid enhancing technologies;
- Encourage modernization efforts to include security in the underlying control network as attacks on the power grid are easier than attacking the hardened security of the data center;
- Encourage the use of energy efficient ICT equipment in data centers to maximize available energy; and
- Streamline regulatory requirements to facilitate modernization and addition of diverse energy sources – including lower and no-carbon sources – to the grid.

### **III. STRENGTHENING U.S. CYBERSECURITY WITH AI**

Although cyber threats are constantly evolving, the increasingly sophisticated threats introduced by AI necessitate a paradigm shift in how organizations approach cybersecurity. Cisco is integrating AI across the network, cloud, and endpoints to provide comprehensive protection by improving threat detection, prediction, and response. This approach not only assists security teams by simplifying management and improving outcomes but also augments human insight, allowing cybersecurity teams to operate at machine speed and focus on critical tasks.

#### *A. Support deployment of AI-enabled cybersecurity to address growing threat landscape*

In the face of an increasingly complex cyber threat landscape, AI-enabled cybersecurity solutions will be critical for protecting U.S. economic and national security interests. AI can quickly analyze vast amounts of data, enabling the identification of anomalies that may signal a cyberattack or the detection of malicious code. Moreover, by automating incident response, AI can also significantly reduce response times and tailor security policies to specific organizational needs. However, the future of AI-enabled cybersecurity lies in its ability to help organizations evolve from reactive remediation to proactive prediction and prevention. With the analytical power of AI, cybersecurity tools can identify potential vulnerabilities and forecast cyberattack patterns so that an organization can strengthen its cybersecurity posture before attacks occur.

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<sup>23</sup> Executive Order No. 14154, *Unleashing American Energy*, (Jan. 20 2025), <https://www.whitehouse.gov/presidential-actions/2025/01/unleashing-american-energy/>.



Policymakers can facilitate these advancements by promoting an innovation-friendly regulatory environment that gives cyber defenders the flexibility to address dynamic cyber threats. To ensure that AI is integrated into cybersecurity solutions in a secure and resilient manner, the U.S. Government should:

- Collaborate with the private sector to develop guidance and encourage uptake of internationally recognized standards that define how AI systems and AI-powered technologies are securely deployed; and
- Ensure that AI systems embedded into cybersecurity solutions are not considered high-risk and subject to restrictive regulatory requirements that inhibit the ability of companies to innovate to address emerging threats.

*B. Leverage generative AI to enhance cybersecurity and expand cyber workforce capabilities*

Generative AI (GenAI) is playing an important role in modernizing cybersecurity operations and expanding workforce capabilities. GenAI enhances system usability and democratizes access to advanced security tools, simplifying complex cybersecurity management tasks through natural language processing. For example, GenAI embedded in Cisco's AI Assistant Firewall allows users to configure firewalls with natural language, eliminating the need for technical command-line syntax. This innovation not only lowers the barrier to entry for non-experts but also enhances the efficiency of security operations centers (SOCs).<sup>24</sup> Leveraging security systems with intuitive GenAI interfaces can help expand the pool of cybersecurity talent with less extensive training.<sup>25</sup> To realize the benefits of GenAI for cybersecurity, the U.S. Government should:

- Encourage adoption of cybersecurity systems with GenAI interfaces; and
- Promote the use of GenAI to help address shortages of cybersecurity professionals.

*C. Promote regulatory alignment and compatibility to boost adoption of AI-enabled security*

Regulatory alignment across markets is essential for the widespread and trustworthy adoption of AI, particularly in critical sectors like cybersecurity. Divergent and conflicting regulations create a complex and costly landscape for U.S. businesses, hindering innovation and impeding the deployment of vital AI-powered security solutions. Uniform, risk-based, non-regulatory frameworks - for example those based on international standards - would foster trust, streamline compliance, and enable the rapid deployment of AI-driven cybersecurity tools, ultimately strengthening the U.S. national security posture and promoting a robust, globally competitive AI ecosystem. The U.S. Government should:

- Work with strategic partners and allies to align security certifications to accept internationally recognized schemes for certification, attestation, and verification of products and services to reduce the need for duplication of resources in different countries for these checks; and

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<sup>24</sup> Cisco's SOC Assistant provides comprehensive situation analysis for SOC analysts, correlating intel across the Cisco Security Cloud platform solutions, relaying potential impacts, and providing recommended actions which reduces the time needed for SOC teams to respond to potential threats. See Cisco Systems, Inc., *Cisco Unveils Next-Gen Solutions that Empower Security and Productivity with Generative AI* (Aug. 29, 2023), <https://investor.cisco.com/news/news-details/2023/Cisco-Unveils-Next-Gen-Solutions-that-Empower-Security-and-Productivity-with-Generative-AI/default.aspx>.

<sup>25</sup> Cisco Systems, Inc., *Close the Cybersecurity Workforce Gap with AI Position Paper* (May 3 2024), <https://www.cisco.com/c/en/us/products/collateral/security/xdr/close-cybersecurity-workforce-gap-with-ai.html>.

- Encourage mutual recognition of certification regimes that are substantially similar and discourage the adoption of local requirements that differ from internationally recognized standards.

#### IV. SECURING AI TO SUPPORT INNOVATION

As AI development and deployment evolves across sectors and use cases, robust security is crucial for supporting innovation that protects national security and promotes economic growth. Trustworthy AI encourages adoption by U.S. businesses, underscoring the importance of robust security measures. Safe AI development aligned with human values fosters public trust and supports national security and stability. Without proper safeguards and standards, unintended AI behaviors could harm U.S. businesses and security, highlighting the need for AI guardrails and security research. Cisco is committed to protecting AI transformation by developing solutions like Cisco AI Defense designed to continuously detect and protect against emerging threats in AI applications without compromising speed or safety.

##### *A. Support industry-backed technological solutions to secure AI*

While integration of AI systems will undoubtedly unlock new innovations and efficiencies across the U.S. economy, AI systems can also introduce new risks. AI applications add a new layer to the technology stack in the form of models. Additionally, most organizations use multiple models across public and private clouds. This multi-model and multi-cloud landscape will require government and private sector organizations alike to take a new approach to security. AI-associated risks—like data leakage, data poisoning, training data extraction,<sup>26</sup> prompt injection, and insecure outputs handling—will need to be addressed. Additionally, threats from external actors seeking to exploit vulnerabilities to steal confidential data or otherwise compromise security must be mitigated to reduce risk and liability to U.S. businesses and U.S. national security.<sup>27</sup>

The release of the DeepSeek R1 model exemplifies these challenges, highlighting deficiencies in model security, as demonstrated by Cisco and the University of Pennsylvania’s research on prompt injection vulnerabilities.<sup>28</sup> This research revealed a 100% success rate in jailbreaking the model, underscoring that there are important risk appetite considerations for organizations adopting AI. High performing models do not always equal highly secure models. This highlights the broader issue of commoditized access to model development, which can result in a proliferation of new models with heightened risks and unreliability due to uncertain provenance and unclear data governance. While all AI models can make mistakes, poor security can exacerbate those risks. Developers and deployers of AI models need to have a comprehensive understanding of where those weaknesses and vulnerabilities are and a willingness to fix any identified problems.

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<sup>26</sup> Cisco’s AI research has shown that there are simple methods bad actors can use to extract memorized training data from chatbots, which – if replicable at scale – could have widespread security implications for AI models that are trained on sensitive or proprietary data. See Amy Chang, *Extracting Training Data From Chatbots*, Cisco Blogs (Sept. 20 2024) <https://blogs.cisco.com/security/extracting-training-data-from-chatbots>.

<sup>27</sup> Cisco is a leader in conducting threat analysis. More information on the latest AI security threats can be found in Cisco’s inaugural State of AI Security report. See Cisco Systems Inc., *The State of AI Security 2025 Annual Report*, <https://www.cisco.com/c/en/us/products/security/state-of-ai-security.html>.

<sup>28</sup> Gopal Devarajan, *Evaluating Security Risk in DeepSeek and Other Frontier Reasoning Models*, Cisco Blogs (Oct. 17 2024), <https://blogs.cisco.com/security/evaluating-security-risk-in-deepseek-and-other-frontier-reasoning-models>



Technological solutions can help mitigate these risks. Innovations like automated red teaming can provide continuous testing and validation of AI models to block adversarial attacks. Real-time protection for AI models, such as through Cisco AI Defense,<sup>29</sup> is critical to securely deploying AI in enterprise environments.

Furthermore, it will be important to ensure that models and applications adhere to existing compliance obligations, a growing challenge as AI-powered application development becomes accessible to a wider audience, including non-traditional developers like healthcare and education professionals. As the pool of developers expands, technology is essential to maintain compliance and monitor the usage of AI models over time, ensuring they remain aligned with security and operational standards. Cisco AI Defense plays a crucial role in this process by providing robust tools and frameworks that help developers adhere to compliance requirements and continuously monitor AI applications. To support the secure development and deployment of AI, the U.S. government should:

- Ensure federal agencies deploying AI define risk measurement and mitigation strategies, aided by adopting the NIST AI Risk Management Framework (RMF), and establish controls for AI application access to reduce these risks;
- Collaborate with the private sector to support pre- and post- deployment testing of AI models and encourage automated red teaming and the validation of AI systems;
- Promote the NIST Adversarial Machine Learning (AML) taxonomy<sup>30</sup>; and
- Work with private sector stakeholders<sup>31</sup> and support U.S. participation in AI standards discussions to ensure U.S. leadership, facilitate market access, and reduce duplicative compliance obligations for U.S. businesses.

#### *B. Work with industry to secure AI models' supply chains*

Development and deployment of open-source models has aided U.S. businesses in accelerating AI adoption and helped developers have access to cutting-edge technology to experiment and innovate. However, all organizations should understand the potential risks and vulnerabilities associated with third-party software, AI models, and data. Threat actors have numerous opportunities to create vulnerabilities in AI models such as through the introduction of corrupted data into training datasets or the injection of malicious code in open-source AI repositories. Already industry and its partners are working together to develop methodologies and technologies to assess and mitigate risks in AI models' supply chains.<sup>32</sup> The U.S. government should:

- Work with stakeholders to evaluate weaknesses in publicly available AI models in the United States and help promote market-based solutions to addressing these risks.

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<sup>29</sup> Cisco AI Defense provides two key elements of AI protection: protecting against the risk of sensitive data exposure from employees using third-party systems and sharing IP or other confidential information with AI tools and protecting the development of AI models through validation identification of vulnerabilities, and application of guardrails. See Jeetu Patel, *Protecting AI so AI Can Improve the World, Safely*, Cisco Blogs (Jan. 15 2025), <https://blogs.cisco.com/news/you-cant-sacrifice-ai-safety-for-ai-speed>.

<sup>30</sup> Cisco Systems, Inc., *AI Safety & Security Taxonomy*, <https://www.cisco.com/site/us/en/learn/topics/artificial-intelligence/ai-safety-security-taxonomy.html> (last visited March 11 2025).

<sup>31</sup> The NIST AML taxonomy was developed in partnership with Cisco, NIST, and the UK AI Security Institute and is a successful example of multi-stakeholder collaboration on standards that supports U.S. businesses.

<sup>32</sup> MITRE, *MITRE and Robust Intelligence Tackle AI Supply Chain Risks* (Sept. 27 2023), <https://www.mitre.org/news-insights/news-release/mitre-and-robust-intelligence-tackle-ai-supply-chain-risks>; and Robust Intelligence, *Robust Intelligence Partners with MITRE to Tackle AI Supply Chain Risks in Open Source Models* (Sept. 27 2023), <https://www.robustintelligence.com/blog-posts/robust-intelligence-partners-with-mitre-to-tackle-ai-supply-chain-risks-in-open-source-models>.

## V. KEY ENABLERS FOR AI LEADERSHIP

Development of leading AI systems and solutions will require key enablers to be put in place to ensure that public and private sector organizations can maximize use of AI to support industry-wide innovation and economic growth.

### *A. Support data privacy and security to build trust in AI and promote adoption*

In Cisco's 2024 Consumer Privacy Survey, respondents noted that awareness and implementation of privacy laws led to enhanced comfort with AI adoption.<sup>33</sup> Strong company privacy practices, combined with reasonable privacy laws, are an important part of creating a strong AI ecosystem in the United States. Cisco supports a reasonable federal privacy law to provide a strong foundation for continued growth and innovation in AI; to ensure a harmonized approach across the country (instead of a patchwork of laws that may be confusing for consumers); and to facilitate U.S. leadership globally on privacy and AI. Cisco appreciates the U.S. Government's interest in seeking an approach to federal privacy legislation that will spur innovation and help the U.S. retain its leadership position in AI.

Americans are more likely to use and benefit from the productivity gains associated with AI if they are confident that there are baseline data privacy protections at the federal level, instead of relying on potentially conflicting state privacy laws. Consistent privacy baselines enhance public trust in AI systems, fostering adoption while mitigating risks like unauthorized data use. Similarly, it would position the U.S. as a leader in responsible AI governance, countering perceptions of lax privacy protections; and consequently, American technology providers will be able to better compete in the global market if foreign customers have greater certainty about the applicability of a federal law that governs use of their data when interacting with AI models and AI-powered services from American companies.

### *B. Enable cross-border data flows to support innovation and market access*

Over the past few years, there has been a growing trend of national governments adopting data localization policies. These policies create barriers to international data flows, hindering U.S. companies in their efforts to innovate, leverage AI to strengthen security, and scale data-based products and services to new markets. In fact, numerous independent studies demonstrate the negative impacts of data localization requirements, including: increasing data storage costs; reducing trade volumes; reducing innovation and competitiveness; increased costs for goods and services; and increased regional compliance costs.<sup>34</sup> These policies may also pose risks to privacy and freedom of expression as authoritarian governments often "see physical access to data centers as an enabler of surveillance and political control."<sup>35</sup> Moreover, data localization mandates can

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<sup>33</sup> Cisco Systems Inc., Cisco 2024 Consumer Privacy Survey <https://www.cisco.com/c/en/us/about/trust-center/consumer-privacy-survey.html>.

<sup>34</sup> Nigel Cory & Luke Dascoli, *How Barriers to Cross-Border Data Flows Are Spreading Globally, What They Cost, and How to Address Them*, Info. Tech. & Innovation Found. (July 2021), <https://www2.itif.org/2021-data-localization.pdf>; Conan French et al., *Data Localization: Costs, Tradeoffs, and Impacts Across the Economy*, Inst. for Int'l Fin. (Dec. 2020), [https://www.iif.com/portals/0/Files/content/Innovation/12\\_22\\_2020\\_data\\_localization.pdf](https://www.iif.com/portals/0/Files/content/Innovation/12_22_2020_data_localization.pdf).

<sup>35</sup> *Id.*

result in siloed data in remote places offering less opportunity for monitoring, detecting, and remediating cyber intrusions. To support U.S. competitiveness, the U.S. Government should:

- Oppose overly restrictive data localization mandates that limit U.S. market access, including by negotiating and enforcing provisions in digital trade chapters with trading partners; and
- Seek digital trade provisions that support cross-border data flows, prevent governments from requiring source code as a condition of market access, and ensure non-discriminatory treatment of digital goods and services.

*C. Consider options to protect curated AI datasets*

AI models are only as good as the data they learn from. Data (personal, enterprise, public) and methods applied to enrich it form the foundation that drives the accuracy, efficiency, and reliability of models. Organizations devote significant resources to creating AI/ML models and training data, which can cost millions of dollars to produce. These datasets are becoming high-value assets for companies. Policymakers should explore what kinds of protections are available for these datasets, including whether intellectual property (IP) protections could and should apply in these situations. Companies are interested in protecting both the data used as inputs and the data that is produced (output) of AI systems, to avoid potential theft in the form of distillation models being trained offshore. The U.S. government should work with stakeholders to come up with ideas to deal with the challenge of how to handle ownership of this data.

*D. Ensure robust intellectual property protections for AI use*

As IP protections for AI are contemplated, the U.S. government should consider the importance of the copyrightability of outputs where AI is leveraged. The U.S. should make the requirements for indicating the use of AI in software code clear and workable. Innovators creating software using GenAI need assurance that their work is appropriately protected under copyright law. A strong legal framework would protect software incorporating AI outputs, thereby encouraging investment and innovation. It is important that those who use AI to accelerate the development of innovative software are confident that they can protect their own work using copyright law. The U.S. Government should:

- Ensure that any evolution of copyright law to accommodate GenAI is reasonable and balanced, allowing for copyrightable GenAI source code when there is sufficient human creativity.

*E. Lead in the development of global AI standards*

Cisco has been a leader in technical standards development since the creation of the internet. As part of our annual investment in research and development, hundreds of Cisco technical experts participate in more than 120 standards development organizations each year. We employ recognized international leaders in standards related to Wi-Fi, security, Internet protocols, optics, software-defined networking, mobility, and numerous other technologies, including AI. The quality of Cisco's contributions is well-recognized within the industry and therefore provides the foundation of many telecommunication standards.

The most important goal of standardization is technical interoperability, which allows web browsers to reach websites, laptops to connect to Wi-Fi, and mobile phones to make calls even though the technology that facilitates each of those connections is provided by many different companies around the world. We encourage colleagues internationally to continue to serve as global proponents for the open, industry-led, market-driven standardization system. Decades of global experience in standardization have taught us that the open, industry-led, consensus-based model of standardization yields the most innovative and pro-competition outcomes.

Given the dynamic pace of innovation, the U.S. Government should support industry-led standards and best practices that allow emerging technologies room to grow. Representation from U.S. industry and government is critical to the development of AI standards that align with American values and support American businesses. To promote policies that enable greater participation from a wider range of U.S. stakeholders in Standards Development Organizations (SDOs), the U.S. Government should:

- Host more international standards meetings in the United States;
- Fund support and simplification of the visa processes to make it easier for standards experts to travel to international standards meetings;
- Elevate the level of U.S. leadership involved in standards-related activities to help increase their visibility;
- Work with strategic partners and allied countries to increase international collaboration and participation in industry-led standardization processes to help avoid the creation of competing blocs that could lead to global fragmentation on standards; and
- Ensure SDOs' voting structures support a fair and transparent process so that outcomes are not decided by one country or region.

*F. Promote digital transformation to support adoption of AI in the public sector*

Digital transformation can yield many benefits, including improved and more agile management of resources, better user experience, higher productivity, and reduced costs. Adoption of cloud computing, data analytics, and IoT technologies, creates an AI-ready ecosystem and facilitates the collection, storage, and processing of data, all of which is essential for training and refining AI algorithms. Access to robust, high-quality, and well-structured data is needed for AI to deliver its full potential in enhancing productivity in the public and private sectors.

While the U.S. private sector has made great strides in digital transformation, the U.S. dropped from ranking 10<sup>th</sup> in 2022 to 19<sup>th</sup> in the 2024 UN E-Government Survey.<sup>36</sup> The U.S. Government should view the advancement of AI as a catalyst to ensure public services are fit for the modern era by leveraging digital technologies. If successful, this would improve citizens' experiences of government services and align with broader policy objectives by making public services more tailored and efficient.

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<sup>36</sup> United Nations, *UN E-Government Survey 2024*, <https://publicadministration.un.org/egovkb/en-us/Reports/UN-E-Government-Survey-2024> (last visited March 11 2025).

Replacement of legacy equipment and pivoting to the cloud will not only ease delivery of services, helping to close gaps in serving all Americans, but will also help address some of the energy demand challenges discussed above. At the same time, this shift will also help improve our nation's security posture against potentially disruptive attacks to critical infrastructure by malicious foreign actors. The discovery that state-sponsored actors had successfully prepositioned footholds in U.S. critical infrastructure systems that deliver electricity and water<sup>37</sup> and accessed sensitive calling data and wiretap records<sup>38</sup> are strong indicators that we face unaddressed security risks from technology that has not been effectively patched or in some cases is too old to secure. To ensure the U.S. Government can securely harness the benefits of AI to make the provision of public services more effective and efficient, it should:

- Encourage the digitalization of the public sector by prioritizing cloud migration and modernization of legacy infrastructure;
- Develop an effective data strategy that breaks down fragmented data sets and encourages interoperability of data to support creation of high-quality data sets for AI; and
- Streamline U.S. Government procurement processes and increase capacity for testing and validation to ensure that U.S. agencies have access to the latest technology and are not procuring older generations of technology due to slow approvals.

*G. Invest in digital skills and workforce development for AI uptake*

AI will drive significant economic shifts, reshaping industries, labor markets, and competitiveness. U.S. companies are interested in strategic planning for this new economy. Workforce reskilling is key to staying competitive and ensuring a smooth transition to new opportunities while traditional workflows are disrupted. Harnessing AI's transformative potential will require cultivating digital skills and an AI-ready workforce.

A report from the Cisco-led AI ICT Workforce Consortium found that 92% of technology jobs analyzed are expected to undergo either high or moderate transformation due to AI advancements.<sup>39</sup> But as Vice President JD Vance rightly highlighted in his remarks at the AI Action Summit, the most promising "immediate applications of AI almost all involve supplementing, not replacing, the work being done by Americans." Using AI to augment and strengthen the U.S. workforce will provide new ways to address critical workforce shortages as AI can help ease the transition of workers to high-demand industries. The U.S. is facing a shortfall of 265,000 cybersecurity professionals.<sup>40</sup> Cisco and other tech companies have made significant investments in training workers and students with employer-recognized certifications for high-demand jobs, including cybersecurity. In Fiscal Year 2024, Cisco's Networking Academy trained over 315,000 students in the U.S. and has trained over 2.6 million students throughout the country since the program's inception in 1997. Cisco has also set the goal of training 25 million people on

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<sup>37</sup> Cybersecurity and Infrastructure Security Agency, *PRC State-Sponsored Actors Compromise and Maintain Persistent Access to U.S. Critical Infrastructure* (Feb. 7 2024), <https://www.cisa.gov/news-events/cybersecurity-advisories/aa24-038a>.

<sup>38</sup> John Sakellariadis & Maggie Mill-linked hackers stole wiretap data from telcos, *FBI and CISA say* (Nov. 13 2024), <https://www.politico.com/news/2024/11/13/china-hackers-wiretap-data-telcos-00189445>.

<sup>39</sup> Cisco Systems, Inc., *AI-Enabled ICT Workforce Consortium*, <https://www.cisco.com/c/m/ai-enabled-ict-workforce-consortium.html> (last visited March 11 2025).

<sup>40</sup> CyberSeek, *CyberSeek Highlights Persistent Cybersecurity Skills Shortage Despite Hiring Stabilization*, PR Newswire (Sept. 20 2023), <https://www.prnewswire.com/news-releases/cyberseek-highlights-persistent-cybersecurity-skills-shortage-despite-hiring-stabilization-302275849.html>.

digital and cyber skills through the Networking Academy by 2032.<sup>41</sup> Yet, the skills gap in cybersecurity persists – and is likely to persist into the future unless we expand the pool of talent from which these roles can be filled.

The U.S. Government and private sector must work together to bridge the gap so that all workers can harness the power of AI and improve productivity. To that end, we support the Administration’s efforts to train the U.S. workforce to ensure that American workers are prepared to integrate, manage, and interact with AI-enabled tools across all sectors. Cisco further recommends that the U.S. Government:

- Research AI’s impact on key industry workforces and share best practices to help companies of all sizes proactively upskill their workforce;
- Initiate public-private partnerships that leverage the technical expertise and insights into workforce demands of the private sector<sup>42</sup>;
- Encourage Congress to reauthorize the Workforce Innovation and Opportunity Act that includes a specific percentage of funds to be dedicated towards worker training programs;
- Encourage and enshrine skills-based hiring practices in both the private and public sectors;
- Extend eligibility for federally subsidized grants and loans, like Pell Grants, to students pursuing short-term training to help them keep pace with rapid AI advancements; and
- Encourage the Department of Labor to recognize non-traditional education programs as valid opportunities for worker development under the National Apprenticeship Program.

## VI. SUMMARY OF RECOMMENDATIONS

This document provides a wide range of recommendations, summarized below, to support U.S. businesses in developing and deploying AI and to ensure the U.S. maintains its global leadership in this emerging technology. Our recommendations reflect Cisco’s expertise from working closely with our customers to support their AI journey, the development and evolution of our AI Governance program, which includes Cisco’s Responsible AI Framework, and our decades of research and development.

**Infrastructure:** We encourage the U.S. government to promote policies that enable investment in data center and broadband infrastructure, the development of future-proofed networks, and affordable and reliable energy. To this end, the U.S. Government should:

- Encourage strategic investments in new data center capacity and modernization of legacy data centers;
- Promote best practices for building AI-ready networks;
- Ensure federal broadband investments are implemented efficiently and quickly;
- Work with permitting agencies to streamline regulatory requirements; and
- Encourage the use of energy efficient ICT equipment in data centers.

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<sup>41</sup>Cisco Systems, Inc., *Cisco to Empower 25 Million People with Digital Skills Over the Next 10 Years* (Oct. 18 2022), <https://newsroom.cisco.com/c/r/newsroom/en/us/a/y2022/m10/cisco-networking-academy-25th-anniversary.html>.

<sup>42</sup> For example, Cisco’s Networking Academy has successfully partnered with federal, state, and local agencies to deliver one of the longest-standing IT skills-to-jobs programs in the world. In the U.S. alone, the Cisco Networking Academy has partnered with a wide range of public and private organizations such as the Department of Labor, the National Collegiate Cyber Defense Competition, Cyber Florida, Wichita State University, and more.



***AI and Cybersecurity:*** To enable the U.S. private sector to innovate with AI to address the dynamic cyber threat environment, the U.S. Government should:

- Collaborate with the private sector to develop guidance and encourage uptake of internationally recognized standards;
- Ensure that AI systems embedded into cybersecurity solutions are not considered high-risk;
- Encourage adoption of cybersecurity systems with GenAI interfaces and promote the use of GenAI to help address shortages of cybersecurity professionals; and
- Encourage mutual recognition of certification regimes that are substantially similar.

***Securing AI:*** To support the secure development and deployment of AI in a way that minimizes risks for U.S. businesses and national security, the U.S. Government should:

- Ensure federal agencies deploying AI define how to measure and mitigate risks with AI;
- Collaborate with the private sector to support pre- and post- deployment testing of AI models and encourage automated red teaming and the validation of AI systems;
- Work with NIST to promote the NIST AML taxonomy; and
- Work with stakeholders to evaluate weaknesses in publicly available AI models and promote market-based solutions to addressing these risks.

***Implementing Key AI Enablers:*** To ensure the United States has the correct tech and workforce readiness for AI, the U.S. Government should:

- Support a reasonable federal privacy law in the United States;
- Seek digital trade provisions that support cross-border data flows, prevent governments from requiring source code as a condition of market access, and ensure non-discriminatory treatment of digital goods and services;
- Consider options to protect curated datasets for AI;
- Support robust IP protections necessary for secure development of AI;
- Facilitate U.S. leadership in the development of global AI standards;
- Promote digital transformation to support adoption of AI in the public sector; and
- Work with private sector partners to invest in digital skills and workforce development.

## **VII. CONCLUSION**

Cisco appreciates the opportunity to assist OSTP in identifying the necessary steps for advancing U.S. leadership in AI. It is imperative to balance a supportive policy environment that allows the AI ecosystem and innovation to flourish while minimizing risks that might inhibit adoption or undermine acceptance of this important technology. Cisco looks forward to working with OSTP to build a prosperous and secure AI ecosystem that supports American businesses and strengthens U.S. economic competitiveness and national security.